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Exploring Optimization-based MRI and Image Reconstruction Methods with PyTorch

Abstract

Method	Dataset	MSE	PSNR	SSIM
Zero-filled FFT (baseline)	MRI	0.01005	19.98 dB	—
Gaussian Kernel	MRI	0.00529	22.78 dB	—
Warm-Start U-Net	MRI	0	137.75 dB*	1
SRCNN	MRI		37.74 dB	0.9534
Kernel + Residual CNN	MRI	0.00248	26.06 dB	0.4153
Zero-filled FFT + Residual CNN	MRI	0.00048	33.20 dB	0.6758
Kernel + Diffusion	MRI	0.00069	31.61 dB	0.7796
Zero-filled FFT + Diffusion	MRI	0.00077	31.12 dB	0.762
SSDiffRecon	MRI	0.002	26.12 dB	0.427
Zero-filled FFT + Residual CNN	Natural Images	0.00038	34.18 dB	0.9104
Zero-filled FFT + Diffusion	Natural Images	0.0045	23.46 dB	0.4564
MFSR-GAN	**Vid4**		**38.66 dB**	**0.9729**

Table 1. Summary table of success metrics across all models (optimized over MSE).

This project evaluates image reconstruction from undersampled k-space measurements, motivated by the need for accelerated MRI acquisition as well as general image reconstruction. We compare various methods, including kernel-based basis functions, Residual CNNs, U-Net, SRCNN, and Diffusion models, across MRI (UPenn-GBM), natural images (Oxford-IIIT Pet), and video (VID4) datasets. Zero-filled FFT provided a strong baseline throughout the project, while zero-filled FFT combined with a residual CNN achieved the best multi-slice MRI results (PSNR 33.20 dB), while multi-frame methods (MFSR-GAN) yielded the highest overall results (38.66 dB PSNR) by leveraging temporal context. We also implemented a self-supervised approach (SSDiffRecon) for reconstruction in environments without ground truth labels.

I. Introduction

This project was initially driven by the clinical and computational hurdles associated with undersampled MRI data. MRI scanners don't measure pixels directly, but instead measure values in frequency space, known as k-space. K-space contains the Fourier coefficients of the image. An image is ideally recovered by applying an inverse Fourier transform to a fully sampled k-space. However, to reduce scan time, MRI data is undersampled, meaning that only a subset of k-space values are collected and some information is missing. As a result, reconstruction becomes an optimization problem, where we try to find an image whose Fourier transform matches the measured k-space data at the sampled locations.

We qualitatively assess success via visual comparison between the ground truth image and our reconstructions. Success was quantitatively measured by two main metrics: MSE (mean squared error) and SSIM (structural similarity index). MSE measures the squared difference between the ground truth and the reconstruction across every pixel, while SSIM compares luminance, contrast, and structure, more similar to human perception of an image than MSE. We optimize against both MSE and SSIM to determine which provides more robust results. We also use PSNR (peak signal-to-noise ratio) as a measure of success, which is a logarithmic scale of the max possible difference squared divided by MSE.

We used kernel-based methods, neural networks, and diffusion models for MRI reconstruction; however, without knowledge of which image sections were most significant medically, we were unable to develop accurate benchmarks for actual MRI diagnostic use. Following this, we broadened the scope of the project to exploring general image reconstruction and deblurring techniques. This included using various neural networks, diffusion models, and multi frame CNN methods.

II. Initial Implementation

Rather than optimizing over all possible images in pixel space, we represent the image as a linear combination of kernel basis functions. The optimization variables are the kernel coefficients $\alpha \in \mathbb{R}^n$, which define the image through $x = K\alpha$, where K is a matrix whose columns are kernel functions evaluated on the image grid. We optimize these coefficients by minimizing the discrepancy between the predicted and observed k-space measurements, ensuring consistency with the measured data while restricting the solution to the kernel image space.

We represent the MRI model by the linear operator $A = MF$, where F denotes the discrete Fourier transform mapping the image to k-space and M is a diagonal matrix with entries 0 or 1 where a value of 1 means the frequency is being measured and a value of 0 means it is not measured. Given observed undersampled k-space data y_{obs}, we solve for α such that $\min_{\alpha \in \mathbb{R}^n} ||A(K\alpha) - y_{obs}||_2^2$

Direct pixel-space reconstruction for undersampled MRI is ill-conditioned because A is not injective, so many distinct images fit the same measured data. This means optimization over all pixel values is unstable due to the unobservable directions. To address this, we restrict the reconstruction to a structured hypothesis {class} defined by kernel basis functions. This removes directions that cannot be reliably inferred from the measurements.

We used data from the UPenn-GBM Cancer Imaging Archive. Data is open-source and can be accessed with the NBIA data retrieval system. This dataset includes full stacks of 2D MRI images: after ordering the images in the order they were taken, we selected a scan in the middle to represent the full stack. We normalized image sizes to minimize noise from misaligned pictures.

We convert the kernel basis from 2D into a frequency in k-space form. To simulate the under-sampling of frequencies that is typical in MRI, we 'mask' the data, adding a filter to only show low-frequency waves. We use an Adam (Adaptive Moment Estimation) optimizer, as it efficiently handles noise.

III. Kernel Reconstruction

To evaluate the effectiveness of our approach, we compared Gaussian and LaPlacian kernels to the zero-filled fast Fourier transform as a base case. Gaussian kernels impose smoothness and locality, which are natural structural assumptions for MRI images and help control high-frequencies. Gaussian kernels also create a more stable objective function under gradient-based optimization. Laplacian kernels have sharper boundaries which may be useful for MRI images, where irregularities can be as small as a few pixels. We use the zero-filled Fourier transform as our base case for comparison of reconstruction quality. This transform assigns a value of '0,' meaning no signal is being measured, to all missing values. Results are reported in Figure 1 and Table 2; the Gaussian reconstruction maintained the highest image quality as measured by PSNR and MSE, improving on the zero-filled FFT baseline, while the LaPlacian had lowest image quality.

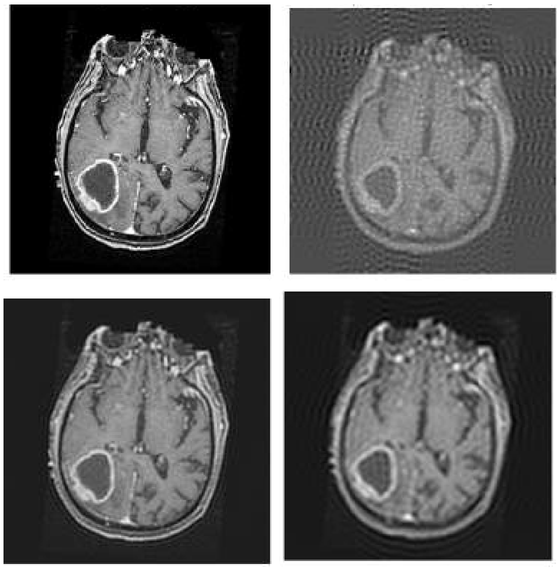


Figure 1. Visual Comparison of Kernel-Based Methods. Top Left: Ground Truth Image. Top Right: LaPlacian Reconstruction. Bottom Left: Gaussian Reconstruction. Bottom Right: Zero-Filled FFT.

Metric	Gaussian	LaPlacian	Zero-Filled FFT
MSE	0.00529	0.0186	0.010049
PSNR	22.78 dB	17.20 dB	19.98 dB

Table 2. Comparison of Kernel-Based Methods by PSNR and MSE

IV. Neural Networks

We trained a neural network to map an undersampled image obtained from inverse FFT of k-space to the ground truth image, formulated as $f_\theta(x_u) = x_t$ where x_u is the undersampled image and x_t is the ground truth image.

An MRI series of 152 slices was loaded and the same slice that was used for the kernel reconstruction (slice 61) was held out from the training set. Then, we undersampled the remaining 119 images and trained a small residual CNN on the undersampled images. The held out slice had a baseline PSNR of 20.0367 dB. When testing on the same slice, the neural network reconstruction had a PSNR of 20.5625 dB for an improvement of 0.5258 dB. Thus, the neural network does generalize and improves the baseline, the zero-filled FFT reconstruction, and the LaPlacian reconstruction, but it is very minimal. Figure 2 shows a visual comparison to the ground truth image.

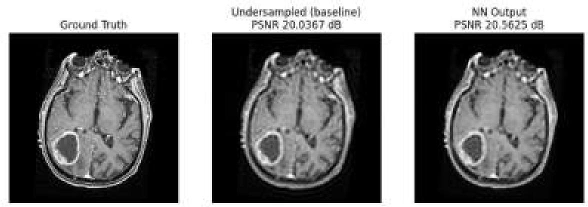


Figure 2. Visual comparison of neural network reconstruction.

Warm-Start Neural Network

We implemented a warm-start neural network approach, using the output from the Gaussian kernel reconstruction as the training data for the neural network. We implemented key changes with this experiment: (1) using a U-Net architecture, which uses convolutional layers of encoder neurons and decoder neurons and is optimized for medical imaging purposes, (2) using a different image to test robustness of the model, since so far we've only used a single image, (3) using SSIM as an additional success metric, which better quantifies visual differences.

This approach, shown in Figure 3, improved MSE by 0.011, PSNR by 118.170 dB, and SSIM by 0.395. It perfectly recreated the target image though we need to train it on more data to make it useful in medical applications. The Gaussian reconstruction (using the new image) had a PSNR of 19.581 dB, MSE = 0.011, and SSIM = 0.605, while the warm-start neural network had a PSNR of 137.75, MSE = 0, and SSIM = 1.0, meaning the neural network reconstructed image completely matched the target image. This suggests this approach would do well on multiple slices, which we attempt in Section V. The warm-start neural network uses significantly lower memory than the original NN implementation.

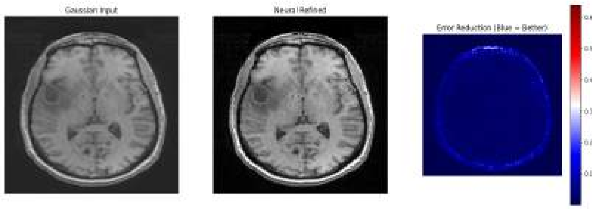


Figure 3. Warm-Start Neural Network Reconstruction using single-slice.

SCRNN

We also implemented a Super-Resolution Convolutional Neural Network (SRCNN) as a refinement stage. SRCNN is a simple but effective three-layer convolutional neural network originally proposed by Dong et al. (2015) for single-image super-resolution. The architecture consists of three layers. The first is a 9×9 convolution with 64 filters. This layer extracts 9×9 patches from the image and projects them into a 64-dimensional feature space. Each of the 64 filters represents a specific visual element such as an edge, a corner, or a texture which allows the network to understand the local structure of the input. The second layer is a 5×5 convolution with 32 filters that learns a nonlinear mapping between low-quality and high-quality feature representations.The final layer is a 5×5 convolution with a single output channel that reconstructs the final image. We show improvements in SSIM and PSNR in Table 3 and visual comparisons in Figure 4.

Metric	Gaussian	SRCNN
SSIM	0.8786	0.9534
PSNR	28.78dB	37.74dB

Table 3. Comparison of SRCNN Metrics to Gaussian Reconstruction

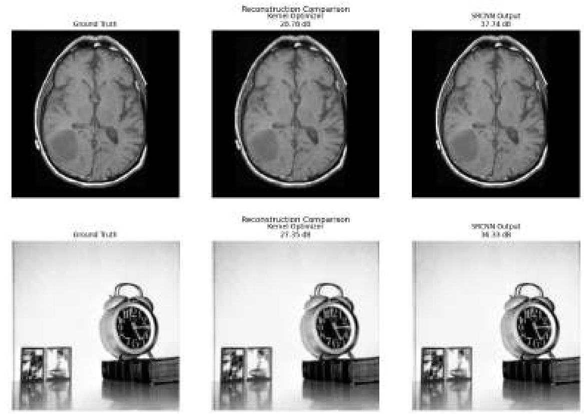


Figure 4: Comparison between ground truth (left), kernel output (middle) and SRCNN output (right), shown with an MRI image (top) and normal image (bottom).

V. Multi-Slice Training

We now pivot to testing and training across multiple slices. We use the same UPENN-GBM collection, evaluating on many training slices rather than just one. We add a TV regularizer to the optimization problem to enforce sharp edges with randomness: $min_\alpha ||A(K\alpha) - y_{obs}||^2 + \lambda ||K\alpha||^2$, where A=MF, which is the undersampling mask, M, multiplied by the Fourier transform k-space data.

The specific hyperparameters optimized were kernel width, number of kernels, and total variation (TV) weight.

To evaluate a candidate set of hyperparameters, we reconstructed multiple training slices and computed reconstruction metrics for each slice. These metrics were then averaged across slices to obtain a single performance score for that hyperparameter configuration. We performed hyperparameter optimization using two different objective functions: mean squared error (MSE) minimization and structural similarity index (SSIM) maximization. This allowed us to compare how optimizing for pixel-wise accuracy (MSE) versus perceptual quality (SSIM) impacts reconstruction performance.

With the updated kernel hyperparameters, we compared this to the baseline reconstruction of zero-filled FFT for a test slice optimized for MSE, shown in Figure 5.

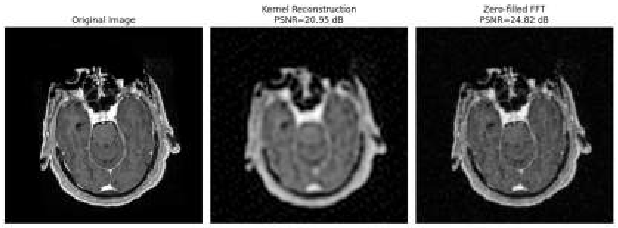


Figure 5. Multi-Slice Reconstruction (MSE) with ground truth (left), kernel (middle) and zero-filled FFT (right).

Optimizing over MSE showed that the multi-slice Gaussian reconstruction returned a lower PSNR (20.98 dB) compared to the zero-filled FFT baseline (24.82 dB). A similar pattern emerged when optimizing over SSIM (shown in Figure 6 and Table 4), as the baseline maintained a higher SSIM, lower MSE, and higher PSNR relative to the Gaussian and LaPlacian reconstructions.

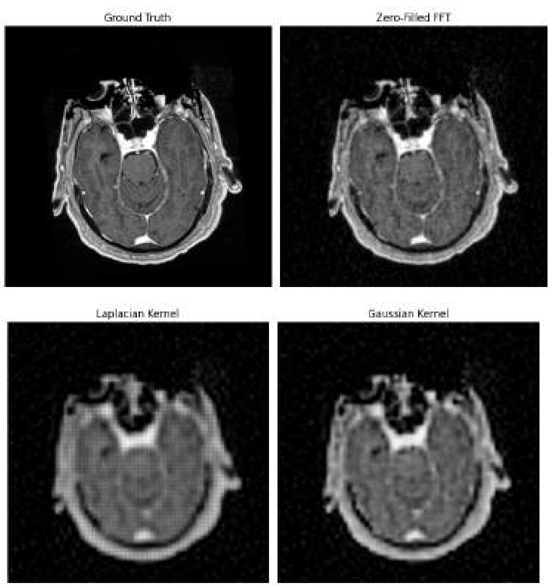


Figure 6. Multi-Slice Reconstruction (SSIM) with ground truth (top left), zero-filled FFT (top right), LaPlacian (bottom left) and Gaussian (bottom right).

Method	MSE	PSNR	SSIM
Zero-Filled FFT	0.002570	25.90 dB	0.584
LaPlacian	0.008532	20.69 dB	0.482
Gaussian	0.007222	21.41 dB	0.511

Table 4. Success Metrics for Multi-Slice Reconstruction (SSIM).

Residual CNN on Multiple Slices

We next used a CNN to act as a refinement step on top of the initial reconstruction method. The basic setup is: $x_{output} = x_{input} + f(x_{input})$, where x_{input} is either the kernel reconstruction or the zero-filled reconstruction and f is the learned residual correction. Essentially, the network doesn't build the entire image from scratch. It cannot change the measured k-space from the original image. It can only refine the improvements the kernel already made.

The residual CNN was tested on MSE-optimized and SSIM-optimized kernel reconstructions and the zero-filled FFT baseline.

Method	Test MSE	Test PSNR	Test SSIM
Kernel Reconstruction & Residual CNN (MSE)	0.001844	27.34 dB	0.634
Kernel Reconstruction & Residual CNN (SSIM)	0.002288	26.41 dB	0.5849
Zero-filled FFT Reconstruction & Residual CNN	0.000569	32.67 dB	0.6904

Table 5. Success Metrics for Residual CNN Corrector + Kernel and Baseline

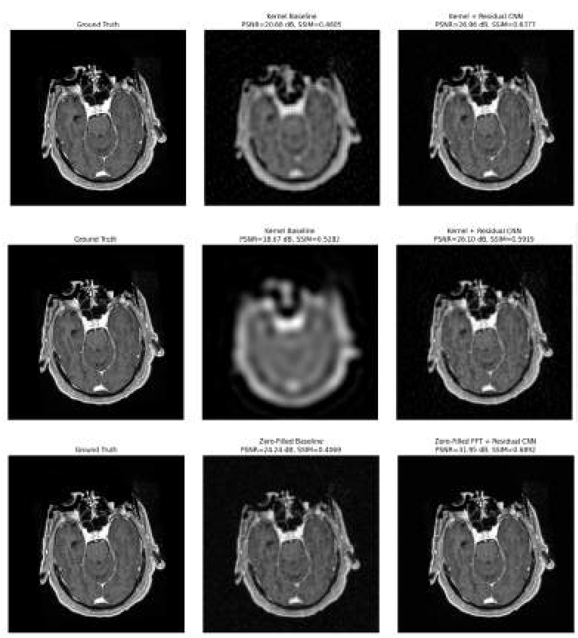


Figure 7. Visual Comparison of Residual CNN-Corrector applied to MSE-optimized Kernel Reconstruction (top) and SSIM-optimized Kernel Reconstruction (middle) with baseline reconstruction (bottom)

The MSE-optimized kernel reconstruction performed better than the SSIM-optimized version after residual CNN refinement, with lower error, higher PSNR, and higher SSIM. This indicates that the SSIM did not provide a better initialization for the CNN here. Instead, the lower-error MSE-optimized reconstruction gave the residual CNN a stronger starting point. Overall, the zero-filled FFT & residual CNN still performed best across all metrics, showing that zero-filled FFT is the most effective baseline.

VI. Diffusion Models

We next moved towards a diffusion model because of its widespread use in image reconstruction. Our setup was similar to the residual CNN, correcting the kernel and baseline reconstructions rather than reconstructing purely from k-space. First, we conditioned the diffusion model on the initial reconstruction (kernel or zero-filled FFT), then it iteratively corrected the areas of the image that were not measured k-space data (i.e., “real” parts of the image).

We again performed the diffusion model correction on the kernel reconstruction and the zero-filled FFT correction as a comparison. The evaluation metrics were MSE, PSNR, and SSIM. See Table 6 for results and Figure 8 for visual comparison. The combination of the kernel reconstruction and diffusion model outperformed the baseline approach with significantly lower runtime.

	Test MSE	Test PSNR	Test SSIM
Kernel Reconstruction & Diffusion Model (MSE)	0.000753	31.23 dB	0.7662
Kernel Reconstruction & Diffusion Model (SSIM)	0.000698	31.56 dB	0.7812
Zero-filled FFT Reconstruction & Diffusion Model Correction	0.000778	31.09 dB	0.7625

Table 6. Success Metrics for Diffusion Corrector + Kernel and Baseline

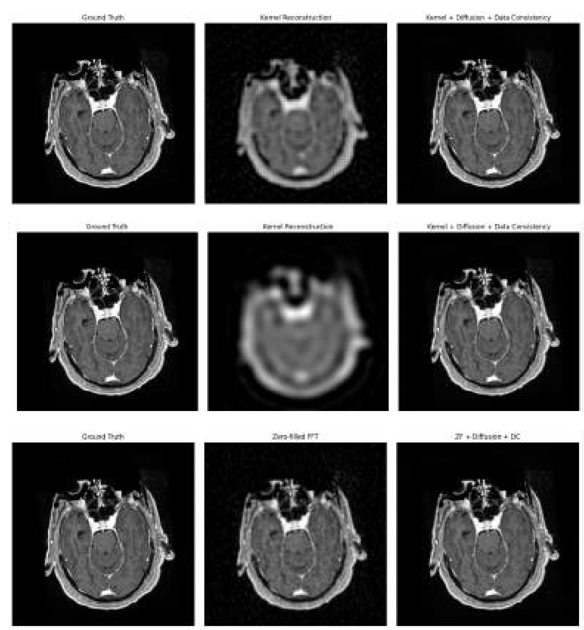


Figure 8. Visual Comparison of Diffusion-Corrector applied to MSE-optimized Kernel Reconstruction (top) and SSIM-optimized Kernel Reconstruction (middle) with baseline reconstruction (bottom)

Overall, the diffusion model improved both kernel-based reconstructions and produced the best results when paired with the SSIM-optimized kernel reconstruction. The kernel reconstructions + diffusion approach slightly outperformed the baseline across all three metrics, suggesting that the diffusion model was better able to exploit the kernel initialization than previous methods.

VII. Self-Supervised Diffusion Models

Thus far, our models have been trained on the ground truth images in order to predict the correct reconstruction. This model builds on diffusion models by using self-supervised learning through training solely on the undersampled *k-space* filters, making it more applicable to medical contexts with reduced data availability. Results were less accurate overall than the original diffusion model implementation, which we expect since the model is solving a harder problem. We were not able to match the accuracy from the literature, likely due to our smaller training dataset and shorter training time.

Our model is based on SSDiffRecon (Korkmaz et al. 2024). We used a split k-space masking strategy, where *M* is the undersampled mask, *Mr* is a random sample of *M* used to train the model, and *Mp* is *M / Mr*, used to test model performance. We start with zero-filled FFT and use cross-attention transformers to denoise, implementing data consistency to ensure no measured k-space values are overwritten. Notably, this model uses unrolled denoising blocks instead of the U-Net Architecture typically used in medical applications. We follow the same architecture as Korkmaz, using an Adam optimizer for self-supervised training with betas = (0.9, 0.999), learning rate 0.02, and *Mr* sampled from M using uniform distribution by taking 8% of measured points. We ran 100 epochs with 20 being reserved for the warm-up. We again compare optimizing over MSE to SSIM.

Results
For the MSE-based loss, we report loss = 0.0220, MSE = 0.0031, SSIM = 0.3844, and PSNR = 25.03 dB, while we report loss = 0.0249, MSE = 0.6174, SSIM = 0.4055, and PSNR = 2.09dB for the SSIM-based loss. We see a major drop in MSE when using SSIM-based loss with modest improvements in SSIM. A visual comparison is available in Figure 9.

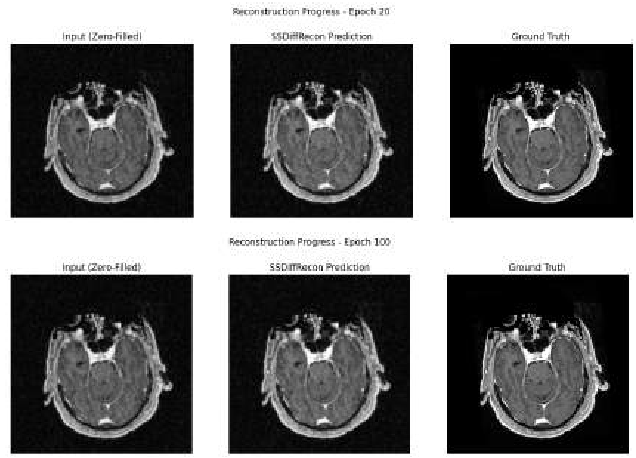


Figure 9. Reconstruction using SSDiffRecon (top: MSE, bottom: SSIM)

VIII. General Images

We broadened the scope of the project to include more general image reconstruction, again comparing optimization over MSE to SSIM. We use the Oxford-IIIT Pet Dataset which includes natural RGB images of animals. Unlike the MRI images, these images contain color, edges, and texture that change the reconstruction assumptions from MRI and are more challenging when sampling from masked Fourier coefficients.

The first step was to determine the type of kernel and kernel hyperparameters. For MSE-optimization, the best configuration was a Laplacian kernel with width of 7.0, TV weight of 1.0e-04, and 1600 kernel basis functions. For SSIM-optimization, the best configuration was a Gaussian kernel with width of 6.68, TV weight of 2.02e-08, and 1764 kernel basis functions. This difference suggests that MSE optimization favors a more localized basis that reduced pixel-wise error, while SSIM optimization favored a smoother Gaussian basis that preserved broader structural similarity.

Notably, this deviates from the MRI reconstruction, where the Gaussian outperformed Laplacian. We also constructed the zero-filled FFT reconstruction as a baseline: optimizing over MSE yielded a mean MSE of 0.003692 and mean PSNR of 24.430 dB, while optimization over SSIM showed a mean MSE of 0.004005, mean PSNR of 24.102 dB, and mean SSIM of 0.6762. We show representative test images in Figure 10.

Figure 10. Ground truth image (left), LaPlacian kernel reconstruction (middle) and zero-filled baseline (right). Optimized over MSE (top) and SSIM (bottom).

Similar to the MRI images, zero-filled FFT also achieved a higher PSNR compared to the kernel reconstruction.

Next, we trained a residual CNN to correct only the directions that were not already from the undersampled Fourier transform. We did this for both the kernel reconstruction and the zero-filled FFT baseline. See success metrics in Table 7 and a visual comparison in Figure 11.

	Kernel & CNN (MSE-optimized)	Kernel & CNN (SSIM-optimized)	Zero-Filled FFT & CNN
MSE	0.000833	0.000807	0.000382
PSNR	30.79 dB	30.93 dB	34.18 dB
SSIM	0.8540	0.8589	0.9104

Table 7. Success Metrics for Residual CNN on Kernel Reconstructions and zero-filled FFT baseline.

Figure 11. Ground truth image (left), initial reconstruction (middle) and reconstruction with CNN (right). MSE-optimized reconstruction (top), SSIM-optimized reconstruction (middle), and zero-filled FFT baseline (bottom).

The residual CNN improved both kernel-based reconstructions, with the SSIM-optimized kernel + CNN performing slightly better than the MSE-optimized kernel + CNN. This suggests that SSIM optimization provided a slightly better structural initialization for CNN refinement. However, the zero-filled FFT warm start again achieved a better result with a higher PSNR and SSIM and lower MSE. This is similar to the MRI reconstruction, showing that even though both images have different assumptions about their overall shape and structure, the same reconstruction patterns show the same results.

Finally, we used a diffusion model to compare the kernel reconstructions and the zero-filled FFT baseline. Success metrics are shown in Table 8 and visual comparisons in Figure 12.

	Kernel & Diffusion (MSE-optimized)	Kernel & Diffusion (SSIM-optimized)	Zero-Filled FFT & Diffusion
MSE	0.004658	0.004358	0.004504
PSNR	23.32 dB	23.61 dB	23.46 dB

SSIM	0.4506	0.4622	0.4564

Table 8. Diffusion Model Success Metrics.

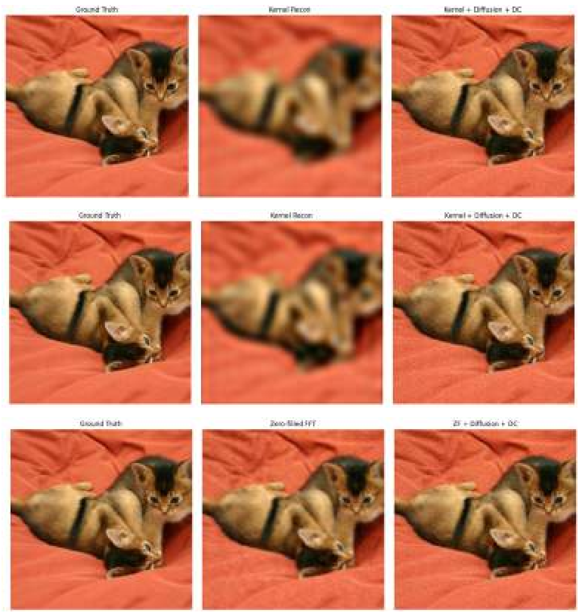


Figure 12. Ground truth image (left), kernel reconstruction (middle) and reconstruction with diffusion model (right). MSE-optimized reconstruction (top), SSIM-optimized reconstruction (middle), and zero-filled FFT baseline (bottom).

The SSIM-optimized kernel + diffusion model performed slightly better than both the MSE-optimized kernel + diffusion model and the zero-filled FFT + diffusion model. However, the improvement was minimal. This shows that, different from the MRI case, the diffusion model did not substantially improve perceptual structure for images.

These results were also much worse than the residual CNN, implying the residual CNN was more effective for refining general image reconstructions. One possible explanation is that the kernel reconstruction already oversmooths natural image details, so the diffusion model has less accurate structure to recover. Overall, the diffusion model was less effective for general images than for MRI images, while SSIM optimization provided only a small improvement over MSE optimization.

XII. Multiframe Reconstruction:

MFCNNs generalize single image CNNs to handle multiple frames and take into account the context of neighboring frames to help strengthen the reconstruction of each individual frame. To retain spatial information, multiple adjacent frames are concatenated along the channel dimension. This allows the network to use extra pixel information from the surrounding sequence to make a more accurate prediction for the middle frames. We implemented two models, the MFCNN (Greaves & Winter, 2016) and the MFSR-GAN (Khan et al. CVPRW 2025). Detailed overviews of these models may be found in Appendix III.

This initial test was done on a standard Vid4 dataset commonly used for multiframe reconstruction. Each frame was downsampled and had noise added to simulate a blurry image.

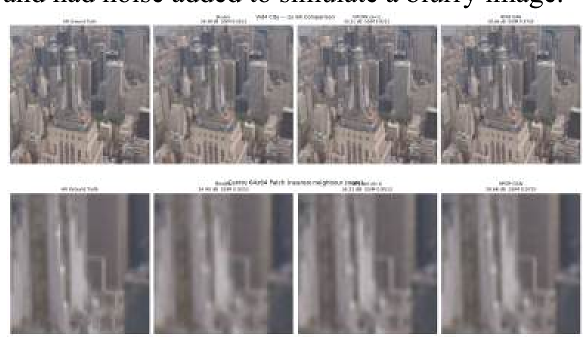


Figure 13. Multi-frame Reconstruction.

Comparing the result in Figure 13 to trying to reconstruct from a singular image, we see that the multiframe method improves PSNR significantly. Training on a singular frame using the CNN, the PSNR only comes to be 33.49 db compared to the 38.66 db that we get when we use multiframe methods.

The primary issue encountered with multiframe methods was compute. While we tried to implement training the model on a larger dataset, the computation time at each level, especially for MFSR GAN, took an incredibly long time to run. This issue was not resolved in time for the end of the semester. Additionally, there were some issues in trying to parse the data from the MRI Dataset to find the specific frames that were neighboring since the ordering did not seem to be uniform.

XII. Conclusion

In practice, physicians would typically be combining multiple low-resolution MRI images into a single high-resolution one using these reconstruction methods: using multiple images or even an entire stack would introduce computational constraints as the filesize increases.

Even our most technically advanced reconstruction model, SSDiffRecon, doesn't match increased accuracy that the researchers reported, likely due to our smaller training set and changes in implementation. Giving the model a more difficult objective to solve (with a smaller k-space mask and a stricter data consistency requirement) makes it more difficult for the model to make changes from the input. Future work may explore using larger datasets that are optimized for MRI reconstruction, such as fastMRI.

Across all methods and both datasets, the most consistent finding was that zero-filled FFT provided a strong baseline, one that kernel reconstruction rarely surpassed. This pattern held for both MRI and natural images, which shows that the structural assumptions kernel basis functions use may be too restrictive when compared to the simplicity of zero-filled FFT. The best strategy throughout was to treat zero-filled FFT as the starting point and apply a learned refinement on top of it, with the residual CNN combination achieving the best multi-slice MRI results at 33.20 dB PSNR.

Taking the project from MRI to natural RGB images using the Oxford-IIIT Pet Dataset showed that these reconstruction patterns generalize across image types and not just MRIs. While the preferred kernel type differed, the relative ordering of methods remained consistent, with zero-filled FFT plus residual CNN doing better than the kernel-based in both types of images.

The strongest absolute results came from multi-frame methods, where MFSR-GAN reached 38.66 dB PSNR which leverages temporal context across adjacent frames. A natural next step would be applying multi-frame reconstruction directly to MRI stacks, where adjacent slices provide similar adjacent contexts. With access to a larger benchmark dataset, and with the diffusion model fully functional, future work could better evaluate where each method sits relative to SOTA reconstruction methods.

For SSDiffRecon specifically, optimizing over MSE vs SSIM did not produce a significant difference in image quality but drastically changed the MSE. When optimizing over SSIM, MSE increased from 0.0031 to 0.6174 and PSNR dropped from 25.03 dB to 2.09 dB, while SSIM improved only from 0.3844 to 0.4055. This suggests that for SSDiffRecon, SSIM-based optimization destabilizes the reconstruction without meaningful perceptual benefit, and MSE-based training is the more reliable objective for this model.

XIII. Appendices

Appendix I. Multi-Slice Hyperparameter Tuning

To evaluate a candidate set, we reconstructed multiple training slices and computed reconstruction metrics for each slice. These metrics were then averaged across slices to obtain a single performance score for that hyperparameter configuration.

Hyperparameter Results

Optimization using MSE

Kernel Type	Sigma	Kernel #	TV Weight	Test MSE	Test PSNR
LaPlacian	4.1017	1600	1.3145e-03	0.009615	20.173 dB
Gaussian	2.5	3844	0.0	0.007886	21.032 dB

Optimization using SSIM

Kernel Type	Sigma	Kernel #	TV Weight	Test MSE	Test PSNR	Test SSIM
LaPlacian	8.57	1936	1.07e-06	0.0110	19.38 dB	0.58
Gaussian	7.09	2411	2.41e-08	0.0121	19.85 dB	0.56

The parameter search showed a tradeoff between MSE and SSIM optimization. MSE optimization selected smaller sigma values and more kernel basis functions. The Gaussian kernel performed slightly better under this objective.

SSIM optimization selected larger sigma values and fewer kernel basis functions. These models had worse MSE and PSNR, but much higher SSIM, with the Laplacian kernel achieving the best SSIM overall.

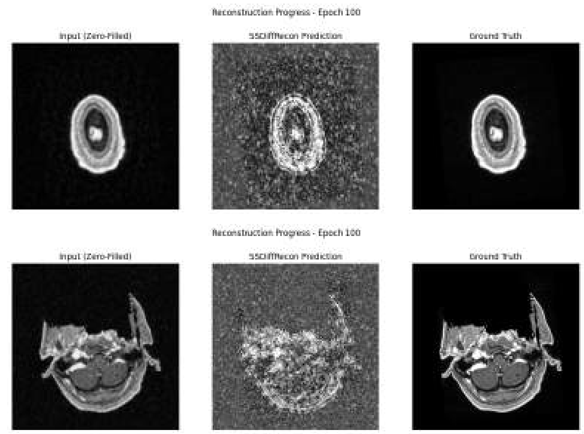
These results demonstrate that the optimization objective strongly affects the type of reconstruction learned. MSE optimization produced more pixel-accurate reconstructions, while SSIM optimization produced reconstructions that better preserved perceived structure of the MRI slice. For MRI images, this distinction matters because a reconstruction can have worse pixel-wise error while still maintaining better anatomical structure.

Appendix II: SSDiffRecon

On the first run, the model immediately hit a noise floor and made few changes beyond the zero-filled input.
In an effort to see the model ‘do something,’ even if reconstruction quality declines, we implemented changes including:

- Sigmoidal activation function: We implemented a sigmoidal activation function to constrain the output to the [0,1] range, preventing extremely bright points in k-space from dominating the image and making model improvements clearer.
- Cosine Annealing Scheduler with warmup: We start with a linear scheduler before switching to cosine annealing so that the model can learn the general shape of the image before dropping the learning rate.
- Frequency Weighted Loss: We implemented focal frequency loss because the model had a lot of trouble initially filling in details - rather, it duplicated the input with minimal changes.

We report frequency-weighted loss of 0.167, MSE = 0.110, SSIM = -0.05, and PSNR = 9.59 dB. Though the loss is relatively low, the near-zero SSIM reflects the model’s difficulty in actually creating an image that is similar to the ground truth from the human eye.



Appendix III: Overview of Multi-Frame Reconstruction Models

MFCNN (Greaves & Winter, 2016)

The architecture begins with a pre-processing stage where all five LR frames in the burst are upsampled to the target HR dimensions using bicubic interpolation. This ensures that the network operates in the high-resolution space from the first layer.

To fuse the information from these frames, the images are concatenated along the channel axis. For a standard RGB burst with a temporal window of 5 frames, this results in an input tensor with 15 channels. This allows the initial convolutional layers to directly learn the pixel-wise differences and sub-pixel shifts between the frames, which helps to recover finer details that might be obscured in a single frame.

MFSR-GAN (Khan et al., CVPRW 2025)

The primary challenge in multi-frame reconstruction is that objects move between frames. MFSR-GAN addresses this using two main mechanisms, reference difference computation and deformable convolutions. Reference difference computation involves selecting a "base" frame and calculating the pixel-wise difference between it and the surrounding reference frames instead of treating all frames as equal, the model. This highlights which areas have changed and need more attention. Unlike standard convulsions which use a fixed grid, deformable convolutions can "offset" their sampling locations, allowing the network to effectively "warp" and align misaligned pixels caused by movement or camera shake before they are fused.

Once the frames are aligned, the model uses channel attention and relativistic GAN loss. Channel attention allows the network to "weight" specific feature maps, prioritizing the most informative frames or spectral features for the final output. The Relativistic Discriminator, in MFSR-GAN, shifts the probability that an input x is real, given by a standard Generative Adversarial Network (GAN). This assumption in the standard GAN often leads to a "vanishing gradient" where the generator stops learning if the discriminator becomes too good. Instead of evaluating an image in isolation, the relativistic discriminator predicts the probability that a real image x_r is relatively "more realistic" than a generated image x_g.

This loss minimizes the divergence between the distribution of the reconstructed images and the ground truth images. By making the discriminator "relative," the generator receives a continuous gradient based on how much it is closing the gap toward the ground truth. Standard losses (like MSE) tend to find the "average" of all possible textures, which results in a blurry, grayed-out effect. The GAN loss forces the model to choose a specific, high-frequency texture that is statistically indistinguishable from the training set, recovering sharper details of the image. The L1 loss measures the average of the absolute differences between the predicted pixel values and the ground truth: $L_1 = \frac{1}{n}\Sigma^{n}_{i=1}|y_i - \hat{y}_i|$